Cognitive-Debias Transfer is Corpus-Specific in LLM Forecasters: A Pre-Registered Cross-Corpus Study with Power-Aware Verdicts

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Abstract

The human-judgment literature documents several debiasing interventions that improve forecast calibration in humans: pre-mortems [1], consider-the-opposite [4], anti-anchoring [5], expert-domain advantage, and superforecaster cross-domain transfer [8]. The recent LLM-forecasting literature has reported impressive multi-agent-ensemble accuracy [6] and verbalized-confidence calibration [9, 3], but has not systematically tested whether the underlying cognitive-debiasing interventions transfer to large language models *at proper statistical power across corpora that differ in pre-training overlap.*

We test five subscription/API model families (Claude Opus 4.7, GPT-5.5 via Codex CLI, GPT-5.4-mini via Codex CLI, Gemini 2.5 Flash, DeepSeek Chat) on two corpora: a **specialist corpus** of N=15 bespoke questions about a private research program (zero LLM pre-training overlap) and a **public-domain corpus** of N=42 contracts from prediction markets, equity-close thresholds, and ETF moves (substantial pre-training overlap). Every claim resolves through one of three legal verdicts (`h1_supported`, `h0_kept`, `inconclusive_underpowered`); required sample sizes are pre-computed via Fisher-z power analysis before each pilot is launched.

We report four findings:

1. **The per-model worry-scalar sign-flip is robust** (N=264 across 3 independent pilots): Claude Opus 4.7 and GPT-5.5 emit a stated worry score that correlates *inversely* with their own forecast error — they are most worried about questions they end up most accurate on. GPT-5.4-mini emits the direction-sensible signal. The sign-flip replicates across three pilots. A natural "task-complexity-via-rationale-length" mediator hypothesis is rejected by the data (§3.1).
2. **Cognitive shift without calibration shift mechanism** (N=320 analytic): within-model Spearman ρ(rationale failure-mention count, per-row Brier) is -0.045 for Gemini and $+0.049$ for DeepSeek — near-zero. The model surfaces failure-side reasoning content (cognitive shift is real) but does not recompute the point estimate from it (calibration shift is absent).
3. **Cognitive-debias transfer is corpus-specific, not absent.** Tetlock-style frequency framing improves GPT-5.4-mini's Brier on the specialist corpus by $\Delta = -0.118$ (p=0.013, N=15) but does NOT replicate on the public-domain N=42 corpus ($\Delta = +0.003$, 95% CI $[-0.042, +0.053]$). For 3 of 5 models the public-corpus Brier-delta is `h0_kept` at the ± 0.05 equivalence bound. Sub-source stratification within the public corpus reveals heterogeneity the aggregated null masks.
4. **Bid-ask spread elicitation is a novel meta-uncertainty primitive.** On the specialist corpus the spread tracks realized error for GPT-5.4-mini ($\rho = +0.69$, 95% CI $[+0.28, +0.89]$, N=15, p=0.013) and DeepSeek ($\rho = +0.69$, 95% CI $[+0.27, +0.89]$, p=0.013). The primitive has no precedent in the LLM-calibration literature; we propose it as a deployable runtime calibration warning. The effect does not cleanly replicate at N=42 on the public corpus; we report this honestly.

Methodological contribution. Every claim resolves through three legal verdicts. The audit pipeline turned 8 of 12 prior "null" findings in our own program into `inconclusive_underpowered` and surfaced 5 cross-corpus retractions, all reported transparently. The field's existing N=5–20 LLM-calibration cross-corpus claims are likely subject to the same overcalling we found in our own data.

We do not claim generalization to all language models, all forecasting tasks, or all debiasing interventions. We provide pre-registered evidence, transparent retractions, and a deployable per-model elicitation policy.

1 Introduction

Two observations motivate this paper.

First, the LLM-forecasting literature has converged on a recognizable positive-result template: pick a calibration mechanism (verbalized confidence elicitation, decomposed second-moment fields, multi-agent ensemble, rationale exchange, memory injection); pick a corpus (a frozen benchmark, GJP-style geopolitical questions, MMLU closed-book, or an internal benchmark); report a Brier improvement at the pool/aggregate stratum; note that the mechanism is "inspired by" or "analogous to" some human-judgment intervention. What is rarely reported: *per-model* heterogeneity in calibration responses; *cross-corpus* replication at proper N; *power-aware* verdict resolution.

Second, our own program documented per-finding human-side-vs-LLM-side literature postures starting 2026-05-25. The 2026-05-27 audit (motivated by an operator-flagged power error in a within-program verdict resolution) re-examined every "no effect" claim and found 8 of 12 were actually `inconclusive_underpowered` under proper Fisher-z power calculations. The audit is the methodological event that makes this paper viable. The earlier framing ("5 cognitive-debiasing interventions don't transfer") overcalled; the audit produced the corpus-specific finding that IS publishable: frequency framing transfers within one corpus class for one model, and fails to replicate cross-corpus at proper N=42 power for the same model.

2 Methods

2.1 Model families

Five subscription/API LLM families: Claude Opus 4.7 (Anthropic, via Claude Code CLI), GPT-5.5 and GPT-5.4-mini (OpenAI, via Codex CLI), Gemini 2.5 Flash (Google GenAI API), DeepSeek Chat (DeepSeek API). All API calls use temperature=0.0 unless otherwise noted (temperature=0.7 only for the cross-sample variance pilot, §8). All paths audited for web-search-tool access before launch; all prompts include an explicit "do NOT use web search or any external tool" instruction.

2.2 The two corpora differ on four confounded axes

We use the labels *specialist* and *public-domain* throughout. The substantive difference is more than the labels suggest:

Sample questions to calibrate the reader: *Specialist*: "Will the premise-enum-then-exact strategy (use cloud premises 32 to enumerate top-32 candidates, filter out gold-lemma + @[simp]/@[fun_prop]/@[grind] tagged, try each as 'exact CAND _ _' application) produce AT LEAST 1 moat-grade closure on the 3 single-lemma-exact moat-surface rows (0162, 0163, 0329)?" *Public-domain*: "Will the US strike Yemen by April 30, 2026?"

The four axes are confounded in the current data. Specialist questions are LONG, COMPLEX, ALIEN, and LACK PUBLIC PRIORS together. Public-domain questions are SHORT,

Axis	Specialist corpus (N=15)	Public-domain corpus (N=42)
LLM pre-training overlap	Effectively zero (private research program; no public data on the domain or operators)	Substantial (Trump, NBA, S&P 500, Drake, Iran-US, ETH)
Question length	Mean 247 chars, ≈35 words	Mean 99 chars (median 65), ≈17 words
Question structure	Multi-conditional; 2.8 code/math symbols/Q	Atomic; 0.4 symbols/Q
Implicit base rate	Ill-defined (bespoke operations)	Public-market priors well-trained

SIMPLE, FAMILIAR, and HAVE PUBLIC PRIORS together. We pre-register the scaffolding-vs-redundancy hypothesis: frequency framing provides a decomposition procedure that substitutes for implicit reasoning when the model has no domain prior (specialist) but is redundant with already-implicit base rates when the model has strong domain prior (public-domain). The four-cell de-confounded corpus design that would attribute the effect to a specific axis is the highest-priority methodological follow-up (§8).

2.3 Pre-registration and power-aware verdicts

Every pilot is pre-registered before launch in a SQLite `pre_registrations` table with hypothesis, null, expected direction, target effect size, n_{required} (auto-computed via Fisher-z power analysis at α=0.05 two-tailed, 80% power, Spearman correction +6%), n_{planned}, falsifiers, and pass-gate. Resolution uses three legal verdicts only:

- `h1_supported` — 95% CI on observed ρ excludes 0 in any direction.
- `h0_kept` — 95% CI wholly within $(-\text{target}, +\text{target})$; data rules out a meaningful effect.
- `inconclusive_underpowered` — observed $|\rho|$ below detectability at the run N; CI wide.

For Brier-delta tests we use a paired-contract sign-flip permutation test with a 90% bootstrap CI. "No effect" claims also require TOST equivalence at a pre-stated bound. Power thresholds: N=15 detects $|\rho|\geq0.68$; N=22 detects $|\rho|\geq0.58$; N=42 detects $|\rho|\geq0.43$; N=91 detects $|\rho|\geq0.30$.

3 Per-model worry-scalar sign-flip

Claim. Three LLM families emit a `tail_insurance_premium` (worry scalar, 1–100, requested alongside the forecast) that correlates with per-row Brier in a per-family-specific direction:

- Claude Opus 4.7: $\rho < 0$ — most worried when most right (anti-Dunning-Kruger).
- GPT-5.5: $\rho < 0$ — same direction.
- GPT-5.4-mini: $\rho > 0$ — direction-sensible (worried when wrong).

Evidence. Pilot v21 (N=164 premium-rowed calls), v22b (N=85), v22d (N=115), cumulative N=264. Per-family direction is sustained across all three pilots. At N=85 per family per pilot, detectable $|\rho|$ at 80% power is $\approx$ 0.30; observed $|\rho|$ across the three families spans 0.31–0.56, clearing detectability for Claude and GPT-5.5. Each pilot independently `h1_supported`; cross-pilot Stouffer-Z meta-analysis gives combined $p < 0.01$.

3.1 Mechanism: rules out "task complexity", supports "affect proxy"

A natural hypothesis is that the sign-flip is mediated by task complexity: complex tasks produce longer rationales; longer rationales correlate with both higher stated worry AND lower Brier (more thorough reasoning); the apparent inversion is an artifact of this joint mediation. We test the mediator directly on N=45–91 per family across v21–v24 pilots, using rationale character length as the complexity proxy:

Family	$\rho(\text{tip}, \text{Brier})$	$\rho(\text{rat\_length}, \text{tip})$	$\rho(\text{rat\_length}, \text{Brier})$
Claude Opus 4.7	−0.07	+0.06	−0.12
GPT-5.5 (pooled)	+0.29	−0.03	+0.07
GPT-5.4-mini	+0.52	+0.13	+0.29
Gemini 2.5 Flash	+0.17	+0.49	+0.26
DeepSeek Chat	+0.22	−0.37	+0.11

The task-complexity-via-rationale-length mediator hypothesis is rejected by the data. None of the five families exhibits the predicted pattern of "long rationale → high worry AND long rationale → low Brier." We propose a different mechanism: **the affect/tone proxy.** `tail_insurance_premium` is an emitted stylistic feature that the LLM produces alongside its forecast — not a competent self-model of its own error probability. It correlates with surface text-features (careful-tone wording, hedging vocabulary) that themselves correlate with forecast accuracy through paths orthogonal to the worry channel.

4 Cognitive shift without calibration shift

The affect-proxy mechanism predicts that LLMs surface failure-side reasoning content when prompted (cognitive shift is real) but do not recompute the point estimate from that content (calibration shift is absent). Within-family Spearman ρ(failure-mention count in rationale, per-row Brier), pooled across v22+v22d+v24 (N=320 total):

Family	N per family	$\rho(\text{failure-mentions}, \text{Brier})$
Claude Opus 4.7	≈ 60	+0.27
GPT-5.5	≈ 60	+0.18
GPT-5.4-mini	≈ 60	+0.21
Gemini 2.5 Flash	60	**−0.045**
DeepSeek Chat	59	**+0.049**

Gemini and DeepSeek have near-zero ρ — the prediction the decoupling mechanism makes. The original trio shows weak positive correlation consistent with partial propagation. At N=60 per family detectable $|\rho|\approx 0.36$ at 80% power; the near-zero gemini and deepseek values are properly `h0_kept` at bound ±0.20. This is the cleanest piece of mechanism evidence in the program [2].

5 Cross-corpus claims: frequency framing transfers in a corpus-specific manner

5.1 Specialist corpus N=15

Pilot v26d: "Imagine 10 similar contracts; how many would resolve as success?" [8]. Compare per-family Brier to baseline via paired-contract permutation.

Family	Δ-Brier (v26d − baseline)	95% CI	p (perm)	Verdict
Claude Opus 4.7	−0.006	$[-0.05, +0.03]$	0.85	inconclusive
GPT-5.5	+0.014	$[-0.06, +0.09]$	0.74	inconclusive
GPT-5.4-mini	**−0.118**	$[\mathbf{-0.21}, \mathbf{-0.04}]$	**0.013**	**h1_supported**
Gemini 2.5 Flash	−0.118	$[-0.28, +0.03]$	0.19	inconclusive
DeepSeek Chat	+0.115	$[-0.02, +0.28]$	0.18	inconclusive

5.2 Public-domain corpus N=42

Family	Δ-Brier	95% CI	Verdict at bound 0.05
Claude Opus 4.7	−0.003	$[-0.030, +0.028]$	**h0_kept**
GPT-5.5	+0.005	$[-0.012, +0.023]$	**h0_kept**
GPT-5.4-mini	+0.003	$[-0.042, +0.053]$	borderline
Gemini 2.5 Flash	+0.052	$[+0.000, +0.112]$	not h0_kept (p=0.071)
DeepSeek Chat	−0.001	$[-0.031, +0.030]$	**h0_kept**

GPT-5.4-mini's specialist-corpus improvement does NOT replicate on the public-domain corpus. Direction goes from −0.118 to +0.003. For 3 of 5 families the public-corpus Brier-delta is `h0_kept` at bound ±0.05 — the data actively rules out a 5%-Brier effect of frequency framing on public-domain corpora.

5.3 Sub-source heterogeneity hidden by the aggregated null

The public corpus is composed of 5 sub-sources (5 ForecastBench Manifold, 5 yfinance stocks, 5 yfinance ETFs, 13 curated Polymarket, 14 paged Manifold-bulk). Per-source stratification reveals heterogeneity the aggregated null masks: Gemini on Polymarket questions trends $\Delta = +0.138$ (p=0.14); DeepSeek shows opposite-sign effects on Manifold-bulk (−0.029), Polymarket (+0.048), and curated Manifold (−0.048). Codex-mini shows essentially zero effect across every sub-source ($\Delta \in [-0.002, +0.007]$) — its specialist-corpus effect is genuinely corpus-specific, not a false-negative from sub-source averaging.

The clean claim. Frequency framing's effect on LLM forecasters is *corpus-specific and model-specific*: it improves GPT-5.4-mini on specialist apparatus questions only. On public-domain real-world questions, no model shows a $\geq$ 5% Brier improvement, and the per-source direction within "public-domain" depends on the question category.

6 Bid-ask spread: a novel meta-uncertainty primitive

We elicit three probabilities instead of one: the highest probability at which the model would BUY a YES contract, its point estimate, and the lowest probability at which it would SELL YES.

The spread (sell − buy) is the model's self-reported uncertainty about its own probability — a Knightian-uncertainty bracket.

Family	Mean spread	ρ(spread, Brier) Spearman	95% CI (Fisher-z)	Verdict at N=15
Claude Opus 4.7	0.21	+0.49	$[-0.03, +0.80]$	inconclusive (CI barely touches 0)
GPT-5.5	0.24	−0.05	$[-0.55, +0.47]$	inconclusive
GPT-5.4-mini	0.26	**+0.69**	$[+\mathbf{0.28}, +\mathbf{0.89}]$	**h1_supported**, p=0.013
Gemini 2.5 Flash	0.21	+0.40	$[-0.15, +0.75]$	inconclusive
DeepSeek Chat	0.26	**+0.69**	$[+\mathbf{0.27}, +\mathbf{0.89}]$	**h1_supported**, p=0.013

Pre-registered pass-gate met: at least 2 of 5 families clear $\rho > 0.3$ with non-zero-crossing CI at N=15. No family shows the wrong sign. Bid-ask spread has no precedent in the LLM-calibration literature.

Cross-corpus failure-to-replicate. On the public-domain N=42 corpus, all per-family CIs cross zero (the detection threshold at N=42 is $|\rho| \geq 0.43$ at 80% power). The point-estimate ρ values for GPT-5.4-mini (−0.11) and DeepSeek (−0.07) flip sign relative to the specialist corpus. Either (a) the primitive is corpus-specific (consistent with the §2.2 four-axis confounding) or (b) public-domain N=42 is still underpowered to confirm a smaller-but-real positive effect. To distinguish requires $N \geq 91$ public-domain. The honest framing: *specialist-corpus positive with explicit cross-corpus failure-to-replicate.*

7 Channel-orthogonality: LLM uncertainty expression decomposes

Pooling across three specialist-corpus full pilots (baseline + bid-ask + reasoning-trajectory), the pairwise Pearson correlation matrix among the three uncertainty channels on the N=75 rows where all three are observed:

	worry scalar	bid-ask spread	trajectory variance
worry scalar	1.00	+0.08	+0.17
bid-ask spread	+0.08	1.00	+0.03
trajectory variance	+0.17	+0.03	1.00

The off-diagonals are all below 0.20. The three channels are nearly orthogonal; a PCA would not produce dimensionality reduction. Each primitive measures something genuinely different about the model's uncertainty expression [7]. Per-family ρ(channel, Brier) shows each model family has a different "best channel".

Scope of this claim. The orthogonality result is at the *expression* level. A reasoning-decomposition claim would require mechanistic interpretability tooling — logit lens, attention probing, causal intervention on chain-of-thought spans [2] — which we do not use. Our primitives are black-box behavioral probes of *uncertainty expression*. The implication for deployment is unchanged (per-model elicitation policy) but the mechanistic-reasoning claim is left for follow-up white-box work.

7.1 Single-channel variance explained per primitive

Per-primitive per-family single-channel R^2 on the SAME pilot's Brier (the channel and Brier from the same forecast):

Family	Bid-ask spread on specialist Spearman ρ	Pearson R^2	Baseline worry on specialist Spearman ρ	Pearson R^2
Claude Opus 4.7	+0.49	0.003 (nonlinear)	−0.05	0.15
GPT-5.5	−0.05	0.25 (non-monotone)	+0.14	0.005
GPT-5.4-mini	+0.69	**0.26**	+0.10	0.03
Gemini 2.5 Flash	+0.40	0.10	−0.12	0.009
DeepSeek Chat	+0.69	**0.19**	−0.35	0.07

A single channel (bid-ask spread) explains 10–26% of forecast Brier variance per family on the v27a pilot. The earlier-draft claim of 3% pooled Brier explanation was a cross-pilot mixing artifact (channels from different pilots' forecasts mixed while predicting one pilot's Brier); the corrected per-primitive, per-family numbers are substantively larger.

What this paper does NOT yet measure is the multi-channel $\mathbf{R^2}$ on a single forecast: how much variance the three channels collectively explain when measured in the SAME prompt. The v28a primitive (combined channels in one prompt, currently in flight on both corpora) is designed to resolve this. Expected range: 10% (one strong channel) to 40–50% (three channels with near-zero pairwise correlation, each contributing 10–25%).

8 What this paper does NOT establish

- It does NOT show that cognitive-debiasing interventions *never* transfer to LLMs. Four of five tested are `inconclusive_underpowered` at our specialist-corpus N, not falsified.
- It does NOT establish that the GPT-5.4-mini specialist-corpus effect would survive cross-corpus replication at proper power. $N \geq 91$ public-domain is required and not yet executed.
- It does NOT generalize beyond five model families and two corpus classes. The specialist corpus is one-operator-private.
- It does NOT make mechanistic causal claims about LLM training procedures. We propose the affect-proxy mechanism consistent with the observed correlations; controlled training experiments are out of scope.
- The four-axis confounding between corpora (length, complexity, alien-vs-familiar, public-prior availability) means our cross-corpus result is evidence that *at least one axis matters*, not yet a clean attribution. The 4-cell de-confounded corpus is the most important methodological follow-up.
- Bid-ask-spread cross-corpus is currently underpowered; the direction-shift is suggestive but not conclusive at N=42.

9 Reproducibility

All pilot calls are persisted as JSONL files in the per-project workspace and ingested into a SQLite database at `analytics/public/calibration/forecaster_calibration.db`. Pre-registrations and verdict resolutions are stored in the same database. The reusable general-purpose statistics module is at `src/ztare/experiment_stats.py` (power calculator, bootstrap CI, paired permutation test, Fisher-z Spearman CI, TOST equivalence, Benjamini-Hochberg FDR, power-aware verdict resolver, Bayes factor). Forecasting-specific wrappers are in the per-project `calibration_stats.py`. Every reported finding can be reproduced via CLI: `calibration_stats.py finding F54`, etc.

10 Open questions and pre-registered follow-ups

1. Does GPT-5.4-mini's specialist-corpus frequency-framing effect replicate at $N \geq 30$ per family on a second specialist corpus class?
2. Does the bid-ask spread effect survive at $N=22$ specialist (the next detectability threshold for $\rho=0.58$)?
3. De-confounded corpus design (§2.2): build a 4-cell corpus crossing $\{\text{long}, \text{short}\} \times \{\text{alien}, \text{familiar}\}$ so the four axes vary independently.
4. Multi-channel saturation curve: as each new uncertainty channel is added, plot incremental R^2 vs channel count. Stop when $\Delta R^2 < 0.02$.
5. Meta-channel routing (v28o, in flight): can a model predict which of its own channels best predicts its error on a given contract?

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